MATH 243 — Probability and Statistics

Random Variables: pmf, pdf, CDF, Expectation, Variance and Chebyshev

Lesson notes · Exam approach · TF practice with solutions · Quick notes (KKS Sir's portion — Probability & ANOVA)

How to Read These Notes

EXAM FOCUS	Exam importance, where it appears in the TF collection
MEMORIZE	Rules, tables and formulas to memorize
TRAP	Common mistakes and exam traps
EXAM-READY ANSWER	Model wording you can write in the exam
NOTE	Extra explanation / clarification
Grey box, monospace	Code, worked tables, derivations and hand-drawn sketches
Centered grey box	Formula to remember / final numerical result
Purple heading / ✓ green	Practice questions / their solutions

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Lesson 9 — Expectation, Variance and Chebyshev's Theorem — E(X), V(X), E[g(X)], aX + b, Chebyshev
Quick Notes — Random Variables — revision bullets

NOTE Page numbers refer to KKS Sir's Lecture 6.pdf (7 pp.), Lecture 7.pdf (8 pp.), Lecture 9.pdf (9 pp.) and Lecture on other distributions...pdf (10 pp. of Walpole, Myers, Myers & Ye, 9th ed.; Chebyshev on p. 1). The exponential distribution is asked in TF but is not in the lectures; it is taught here from the textbook and marked. "TF p." = page of the merged question PDF; "22 batch p." = 2023-24 paper.

Exam Map — What the TF Papers Ask

Session (paper)	Questions answered by this note [lesson]
2023-24 (MATH 243) TF p. 22 batch p.1, 22 batch p.2	Q2(a) $F(x) = 1 - e^{-8x}$: P(wait < 12 minutes) using the CDF and using the pdf (15) [L8] · Q3(a) Chebyshev's theorem; 70 jobs, 1000 applicants, mean 60, sd 6: can a score of 84 get a job? (11) [L9]
2022-23 (MATH 243) TF p. 47, 48	Q1(c) DVD component $f = (1/2000)e^{-x/2000}$: CDF; P(lasts > 1000 h); P(fails before 2000 h) (12) [L8] · Q2(b) Clearance time $Y = 3X - 2$, X with $f = (1/4)e^{-x/4}$: mean and variance of Y (10) [L9] · Q2(c) $f = 6x(1 - x)$: $P(\mu - 2\sigma < X < \mu + 2\sigma)$ compared with Chebyshev (10) [L9]
2020-21 (MATH 245) TF p. 38	Q6(b) Measurement error $f = k(3 - x)^2$, $-1 \le x \le 1$: k; P(X < 1/2); P(|X| > 0.8) (10) [L8] · Q8(a) Hospital stay $Y = X + 4$, $f(x) = 32/(x + 4)^3$, $x > 0$: average stay E(Y) (10) [L9]
2019-20 (MATH 245) TF p. 35	Q7(b) $f(x) = e^{-x}$, $x > 0$: expected value of $g(X) = e^{2X/3}$ (18) [L9] · Q8(b) Fire-station calls, exponential with rate 1.8/day: five parts (15-min call, memoryless, 10-h shift, 4 shifts, 10 % point) (25) [L8]
2018-19 (MATH 245) TF p. 24, 25	Q7(a) Density function and its conditions; $F(x) = x/8$, $x^2/16$ piecewise: pdf, $P(X \ge 1.5)$, $P(X \ge 1 \mid X \le 3)$ (20) [L8] · Q7(b) Lot of 7 (4 good, 3 defective), sample 3: expected number of good components (15) [L9]

EXAM FOCUS 11 questions, 156 marks — 9 of them (121 marks) since 2019-20, so this is the heaviest probability note. Three types repeat: (1) find k / the CDF / a probability from a density; (2) exponential waiting times (2019-20, 2022-23, 2023-24); (3) E[g(X)], mean and variance of aX + b, and Chebyshev (2022-23, 2023-24). Learn the integration steps and the three formulas $E(X) = \int x f$, $V(X) = E(X^2) - \mu^2$, $F(x) = \int f$.

Lesson 7 — Random Variables and the Probability Mass Function

Lec 6 pp. 1–7

1. Random Variable (Lec 6 p. 1)

A **random variable** is a variable whose values are numbers that arise from chance, so that they cannot be predicted exactly in advance. It can be viewed as a function from the sample space to the real numbers. Example: a committee of 2 from 4 male and 3 female teachers; Y = number of males takes the values 2 (M, M), 1 (M, F), 0 (F, F).

Type	Definition	Examples
Discrete	finite or countable set of isolated values	telephone calls per day; under-five children in a family
Continuous	any value in an interval	weight of a six-month-old baby; life of an electric bulb

2. Probability Distribution (Lec 6 p. 2)

A probability distribution associates each value (or class) of a random variable with its probability. Discrete examples: Bernoulli, binomial, Poisson. Continuous examples: uniform, normal, exponential.

Example 1 — black balls: 10 balls, 4 black, 3 drawn without replacement (Lec 6 pp. 2–3)

```
f(x) = 4Cx × 6C(3−x) / 10C3,  10C3 = 120
x     :    0        1        2        3
f(x)  :  20/120   60/120   36/120   4/120     (total 1)
```

Example 2 — number of runs in three tosses (Lec 6 pp. 3–4)

A run is a maximal block of identical consecutive outcomes. HHH, TTT → 1 run; HHT, HTT, THH, TTH → 2 runs; HTH, THT → 3 runs. So P(X = 1) = 2/8, P(X = 2) = 4/8, P(X = 3) = 2/8.

> **NOTE** The slide's outcome table lists THH twice; the eighth outcome is THT (3 runs). The final distribution 2/8, 4/8, 2/8 is correct.

3. The pmf and its Conditions (Lec 6 p. 4)

$$f(x) = P(X = x)$$

$$(1)\ f(x) \geq 0 \quad (2)\ \Sigma f(x) = 1$$

Function	Check	pmf?
(2x − 1)/8, x = 0, 1, 2, 3	sum = (−1 + 1 + 3 + 5)/8 = 1, but f(0) = −1/8 < 0	No
(3x + 6)/21, x = 1, 2	9/21 + 12/21 = 1, all positive	Yes
(x + 1)/16, x = 0, 1, 2, 3	(1 + 2 + 3 + 4)/16 = 10/16 ≠ 1	No

> **NOTE** The slide prints (b) as "(3x=6)/8"; its own sum 9/21 + 12/21 shows the function is (3x + 6)/21.

4. Finding k (Lec 6 pp. 5–7)

X = 0, …, 7 with P = 0, k, 2k, 2k, 3k, k², 2k², 7k² + k. Σ = 10k² + 9k = 1 → (10k − 1)(k + 1) = 0 → k = 1/10 (k = −1 is impossible).

```
P(X ≤ 6) = 1 − P(X = 7) = 1 − (7k² + k) = 1 − 17/100 = 83/100
P(3 < X ≤ 6) = P(4) + P(5) + P(6) = 3k + k² + 2k² = 0.30 + 0.01 + 0.02 = 33/100
```

Geometric-type pmf $f(x) = \alpha(3/4)^x$, x = 0, 1, 2, …: Σ = α/(1 − 3/4) = 4α = 1 → α = 1/4, so $f(x) = (1/4)(3/4)^x$ and P(X ≤ 3) = (1/4)(1 + 3/4 + 9/16 + 27/64) = 175/256. (The slide writes $(1/4)(1/4)^x$ by a slip.)

Exercises of Lecture 6 (p. 7) — answers

pmf	Σ → k	Probabilities asked
kx; x = 1, 2, 3, 4	10k = 1 → k = 1/10	P(X = 3) = 3/10; P(X > 2) = 7/10
k(x² + 1); x = 0, 1, 2, 4	25k = 1 → k = 1/25	P(X = 2) = 5/25 = 0.2; P(X ≥ 2) = 22/25; P(X ≤ 1) = 3/25
k(x + 2); x = −1, 0, 1, 2, 4	16k = 1 → k = 1/16	P(X < 1) = 3/16; P(X ≥ 0) = 15/16

Exam Approach

1. List the possible values of X and find each probability (counting, or by formula).
2. Check f(x) ≥ 0 and Σ f(x) = 1; solve Σ f(x) = 1 for an unknown constant, rejecting negative roots.
3. Add the required f(x); use the complement when it is shorter.

Practice — Lesson 7

Constructed (Lec 6 exercise types).

(a) Two pens are chosen from 3 red and 5 black. Find the distribution of X = number of red pens.

(b) f(x) = c(x + 1)², x = 0, 1, 2. Find c and P(X ≥ 1).

✓ Solution

(a) 8C2 = 28. f(0) = 5C2/28 = 10/28, f(1) = 3 × 5/28 = 15/28, f(2) = 3C2/28 = 3/28 (sum 1).

(b) c(1 + 4 + 9) = 1 → c = 1/14; P(X ≥ 1) = 1 − 1/14 = 13/14.

Lesson 8 — Density Function and Distribution Function (CDF)

Lec 7 pp. 1–8 · exponential: textbook

1. Discrete CDF (Lec 7 pp. 1–3)

$$F(x) = P(X \le x) = \Sigma_{t \le x}\, f(t)$$

F is non-decreasing, $0 \le F(x) \le 1$, $F(-\infty) = 0$, $F(\infty) = 1$, and it is a step function for a discrete X.

```
Black balls:     x < 0: 0 | 0 ≤ x < 1: 20/120 | 1 ≤ x < 2: 80/120 | 2 ≤ x < 3: 116/120 | x ≥ 3:
1
Heads in 3 tosses: x < 0: 0 | [0,1): 1/8 | [1,2): 4/8 | [2,3): 7/8 | x ≥ 3: 1
Fair die:        F(x) = k/6 on [k, k + 1), k = 1, …, 5;  0 for x < 1;  1 for x ≥ 6
Bernoulli(p):    0 for x < 0;  1 − p on [0, 1);  1 for x ≥ 1
```

2. Probability Density Function (Lec 7 pp. 4–5)

EXAM-READY ANSWER A probability density function f(x) of a continuous random variable X is a non-negative function whose total area under the curve (over the whole range of x) equals one; probabilities are areas under the curve. Conditions: (1) $f(x) \ge 0$ for all x; (2) $\int_{-\infty}^{\infty} f(x)\,dx = 1$; then (3) $P(a < X < b) = \int_a^b f(x)\,dx$. For a continuous X, $P(X = a) = 0$, so < and ≤ give the same probability. (2018-19 Q7(a))

Example (Lec 7 pp. 4–5): $f(x) = kx$, $0 < x < 4$: $k[x^2/2]_0^4 = 8k = 1 \to k = 1/8$; $P(1 < X < 2) = [x^2/16]_1^2 = 3/16$; $P(X > 2) = [x^2/16]_2^4 = 3/4$. Example: $f(x) = (2/27)(1 + x)$, $2 < x < 5$: $\int = (2/27)[x + x^2/2]_2^5 = 1$; $P(X < 4) = (2/27)(12 - 4) = 16/27$; $P(3 < X < 4) = (2/27)(12 - 7.5) = 1/3$.

Exercises of Lecture 7 (p. 6) — answers

pdf	k	Probabilities
$k(4 - x^2)$, $0 \le x \le 2$	$\int = 16/3 \to k = 3/16$	$P(X = 1) = 0$; $P(0.5 \le X \le 1.5) = 0.5469$; $P(X \ge 1) = 0.3125$
$k(x^2 + 1)$, $0 \le x \le 3$	$\int = 12 \to k = 1/12$	$P(X \ge 2) = 0.6111$; $P(1 \le X \le 1.5) = 0.1076$; $P(X \le 2) = 0.3889$
kx^2e^{-x}, $x \ge 0$	$\int = 2! \to k = 1/2$	$P(X \le 1) = 1 - 2.5e^{-1} = 0.0803$; $P(1 \le X \le 3) = 0.4965$; $P(X \ge 2) = 5e^{-2} = 0.6767$

3. Continuous CDF (Lec 7 pp. 6–8)

$$F(x) = P(X \le x) = \int_{-\infty}^{x} f(t)\,dt, \quad f(x) = F'(x)$$
$$P(a < X < b) = F(b) - F(a)$$

For $f(x) = (2/27)(1 + x)$ on (2, 5): $F(x) = (2/27)\int_2^x (1 + t)dt = (x^2 + 2x - 8)/27$ on $2 < x < 5$ (0 below, 1 above). $F(3) = 7/27$, $F(4) = 16/27$, $F(4) - F(3) = 9/27 = 1/3 = P(3 < X < 4)$ ✓.

Triangular density $f(x) = x$ on (0, 1], $2 - x$ on (1, 2]: $F(x) = x^2/2$ for $0 < x \le 1$, and $F(x) = \int_0^1 t\,dt + \int_1^x (2 - t)dt = 2x - x^2/2 - 1$ for $1 < x \le 2$; $F = 0$ for $x \le 0$ and 1 for $x > 2$.

TRAP For a piecewise density, the CDF on a later piece must include the whole area of the earlier pieces (the "$\int_0^1 t\,dt = 1/2$" term). Going the other way, the pdf from a piecewise F is F'(x) on each piece.

4. The Exponential Distribution (textbook — asked 2019-20, 2022-23, 2023-24)

$$f(x) = \lambda e^{-\lambda x},\ x \geq 0 \quad (\lambda = \text{rate};\ \text{mean } \beta = 1/\lambda)$$
$$F(x) = 1 - e^{-\lambda x}, \quad P(X > x) = e^{-\lambda x}$$
$$E(X) = 1/\lambda, \quad V(X) = 1/\lambda^2$$

- Memoryless: $P(X > s + t \mid X > s) = e^{-\lambda(s+t)}/e^{-\lambda s} = e^{-\lambda t} = P(X > t)$. The time already waited does not matter.
- If calls arrive as a Poisson process with rate λ per unit time, the waiting time between calls is exponential with the same λ, and "no call in t" = P(Poisson count 0) = $e^{-\lambda t}$.
- Keep the units consistent: change the rate to per hour (or per minute) before substituting t.

Exam Approach

1. Find k from $\int f = 1$ over the stated range.
2. Probability = integral over the requested interval (limits clipped to the range of X).
3. CDF: integrate from the lower end of the range to x, piece by piece; state F for every range of x.
4. Exponential: write λ in the right units, then use $P(X \leq t) = 1 - e^{-\lambda t}$.
5. Conditional: $P(A \mid B) = P(A \cap B)/P(B)$ with both found from F.

Practice — Lesson 8

(a) Actual — 2018-19 Q7(a) (TF p. 24). (b) Actual — 2020-21 Q6(b) (TF p. 38). (c) Actual — 2022-23 Q1(c) (TF p. 47). (d) Actual — 2023-24 Q2(a) (22 batch p. 1). (e) Actual — 2019-20 Q8(b) (TF p. 35).

(a) What is a density function and what conditions must it satisfy? F(x) = 0 (x ≤ 0), x/8 (0 < x < 2), x²/16 (2 ≤ x < 4), 1 (x ≥ 4). (i) Find the pdf of X. (ii) Find P(X ≥ 1.5) and P(X ≥ 1 | X ≤ 3). (20)

(b) The measurement error X has density $f(x) = k(3 - x)^2$, −1 ≤ x ≤ 1. (i) Find k. (ii) P(X < 1/2). (iii) The error is undesirable if its magnitude exceeds 0.8: find that probability. (10)

(c) Time to failure (hours) has $f(x) = (1/2000)e^{-x/2000}$, x ≥ 0. (i) Find the CDF. (ii) P(lasts more than 1000 hours). (iii) P(fails before 2000 hours). (12)

(d) The waiting time X (hours) between speeders has $F(x) = 1 - e^{-8x}$, x ≥ 0. Find the probability of waiting less than 12 minutes, using the CDF and using the pdf. (15)

(e) Calls to a fire station: exponential inter-call times, rate 1.8 per day. (i) P(a call in the next 15 minutes). (ii) No call so far in the shift, 15 minutes left: P(a call in the next 15 minutes). (iii) P(no calls in a 10-hour shift). (iv) P(no calls in four successive shifts). (v) In 10 % of shifts there is a call in the first x hours: find x to one decimal place. (25)

✓ Solution

(a) Definition and conditions as in the EXAM-READY ANSWER of §2.

```
(i)  f(x) = F'(x):  1/8 for 0 < x < 2;  x/8 for 2 ≤ x < 4;  0 elsewhere
     check: ∫0^2 1/8 dx + ∫2^4 x/8 dx = 1/4 + (16 − 4)/16 = 1/4 + 3/4 = 1 ✓
(ii) P(X ≥ 1.5) = 1 − F(1.5) = 1 − 1.5/8 = 13/16 = 0.8125
     P(X ≥ 1 | X ≤ 3) = P(1 ≤ X ≤ 3)/P(X ≤ 3) = [F(3) − F(1)]/F(3)
                      = (9/16 − 1/8)/(9/16) = (7/16)/(9/16) = 7/9 = 0.7778
```

> **NOTE** F jumps from 2/8 = 1/4 to 2²/16 = 1/4 at x = 2, so it is continuous there — the pdf is valid.

(b)

```
(i)   ∫_-1^1 (3 − x)² dx = [−(3 − x)³/3]_-1^1 = (64 − 8)/3 = 56/3  →  k = 3/56
(ii)  P(X < 1/2) = (3/56)[−(3 − x)³/3]_-1^1/2 = (1/56)[64 − 15.625] = 387/448 = 0.8638
(iii) P(|X| > 0.8) = 1 − P(−0.8 ≤ X ≤ 0.8) = 1 − (1/56)[3.8³ − 2.2³]
                   = 1 − (1/56)(54.872 − 10.648) = 1 − 44.224/56 = 0.2103
```

(c) Exponential with mean 2000 h (λ = 1/2000).

```
(i)   F(x) = ∫0^x (1/2000)e^(−t/2000) dt = 1 − e^(−x/2000) for x ≥ 0;  F(x) = 0 for x < 0
(ii)  P(X > 1000) = 1 − F(1000) = e^(−0.5) = 0.6065
(iii) P(X < 2000) = F(2000) = 1 − e^(−1) = 0.6321
```

(d) 12 minutes = 0.2 hour.

```
CDF:  P(X < 0.2) = F(0.2) = 1 − e^(−8×0.2) = 1 − e^(−1.6) = 0.7981
pdf:  f(x) = F'(x) = 8e^(−8x), x ≥ 0
      P(X < 0.2) = ∫0^0.2 8e^(−8x) dx = [−e^(−8x)]0^0.2 = 1 − e^(−1.6) = 0.7981
```

(e) Rate λ = 1.8 per day = 1.8/24 = 0.075 per hour.

```
(i)   15 min = 1/96 day:  P(X ≤ 1/96) = 1 − e^(−1.8/96) = 1 − e^(−0.01875) = 0.0186
(ii)  Memoryless: P(X ≤ s + t | X > s) = 1 − e^(−λt) — the same 0.0186
(iii) 10 h:  P(no call) = P(X > 10) = e^(−0.075×10) = e^(−0.75) = 0.4724
(iv)  four independent shifts: (e^(−0.75))⁴ = e^(−3) = 0.0498
(v)   P(X ≤ x) = 0.10  →  1 − e^(−0.075x) = 0.10  →  x = −ln 0.9 / 0.075 = 1.4048 h ≈ 1.4 h
```

Lesson 9 — Expectation, Variance and Chebyshev's Theorem

Lec 9 pp. 1–5 · other distributions p. 1–2

1. Definitions and Rules (Lec 9 p. 1)

$$E(X) = \Sigma\, x\, f(x) \quad \text{or} \quad \int x\, f(x)\, dx$$
$$E[W(X)] = \Sigma\, W(x)\, f(x) \quad \text{or} \quad \int W(x)\, f(x)\, dx$$
$$V(X) = E[(X-\mu)^2] = E(X^2) - \mu^2, \quad \sigma = \sqrt{V(X)}$$

- $E(c) = c$; $E(cW(X)) = cE(W(X))$; $E[W_1(X) + W_2(X)] = E[W_1(X)] + E[W_2(X)]$.
- Hence $E(aX + b) = aE(X) + b$ and $V(aX + b) = a^2V(X)$ (textbook — the constant shift does not change the spread).

2. Worked Examples (Lec 9 pp. 1–5)

```
x    :  -3     -2     0      1      2
f(x) :  0.10   0.30   0.15   0.40   0.05
E(X)  = -0.3 - 0.6 + 0 + 0.4 + 0.1 = -0.4
E(X²) = 0.9 + 1.2 + 0 + 0.4 + 0.2 = 2.7
V(X)  = 2.7 - (-0.4)² = 2.54,   σ = √2.54 = 1.59
```

- Insurance: gain 2500 w.p. 0.97, 2500 − 25000 = −22500 w.p. 0.03 → E = 2425 − 675 = Tk 1750.
- $f(x) = 2(1 - x)$, $0 < x < 1$: $E(X) = 2\int(x - x^2)dx = 1/3$; $E(X^2) = 2\int(x^2 - x^3)dx = 1/6$; $V = 1/6 - 1/9 = 1/18$.
- Poisson $f(x) = e^{-m}m^x/x!$: $E(X) = e^{-m}\, m\, \Sigma\, m^{x-1}/(x-1)! = e^{-m}\, m\, e^m = m$.
- Markers: 4 good, 3 defective, sample 3: $f(x) = 4Cx \cdot 3C(3-x)/35 = 1/35, 12/35, 18/35, 4/35$ → $E(X) = 60/35 = 12/7 \approx 1.71$.
- Coin game: +5 for HHH or TTT (2/8), −3 otherwise (6/8): $E = 5(2/8) - 3(6/8) = -1$ → loses Tk 1. With 0 for two heads: $E = 5(2/8) + 0(3/8) - 3(3/8) = 1/8$ = Tk 0.125.

> **NOTE** The slide prints the coin-game sum as "5(2/3)"; it is 5 × 2/8, and the answer −1 is right.

3. Chebyshev's Theorem (other distributions p. 1)

$$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - 1/k^2$$
$$\text{equivalently} \quad P(|X - \mu| \geq k\sigma) \leq 1/k^2$$

It holds for every distribution (distribution-free), so the bound is weak: $k = 2$ → at least 3/4 within 2σ; $k = 3$ → at least 8/9. Example 4.27: $\mu = 8$, $\sigma^2 = 9$: $P(-4 < X < 20) = P(8 - 4\cdot3 < X < 8 + 4\cdot3) \geq 15/16$; $P(|X - 8| \geq 6) = P(|X - 8| \geq 2\cdot3) \leq 1/4$.

Proof (textbook, review exercise 4.79)

$$\sigma^2 = \int (x-\mu)^2 f(x)\, dx \;\geq\; \int_{|x-\mu| \geq k\sigma} (x-\mu)^2 f(x)\, dx \qquad \text{(drop the middle part, integrand} \geq 0\text{)}$$
$$\geq \int_{|x-\mu| \geq k\sigma} k^2\sigma^2 f(x)\, dx = k^2\sigma^2\, P(|X-\mu| \geq k\sigma)$$
$$\Rightarrow \; P(|X-\mu| \geq k\sigma) \leq 1/k^2 \;\Rightarrow\; P(|X-\mu| < k\sigma) \geq 1 - 1/k^2 \quad \blacksquare$$

Exam Approach

1. Write $E(X)$ and $E(X^2)$ as sums/integrals over the range; integrate carefully (use $\int_0^\infty x^n e^{-x}dx = n!$).
2. $V(X) = E(X^2) - [E(X)]^2$; for $Y = aX + b$ use $E(Y) = aE(X) + b$, $V(Y) = a^2V(X)$.
3. $E[g(X)] = \int g(x)\, f(x)\, dx$ — do not compute $g(E(X))$.

4. Chebyshev: write the interval as $\mu \pm k\sigma$, find k, then the bound $1 - 1/k^2$ (or $1/k^2$ for the tails).

Practice — Lesson 9

(a) Actual — 2018-19 Q7(b) (TF p. 25). (b) Actual — 2019-20 Q7(b) (TF p. 35). (c) Actual — 2020-21 Q8(a) (TF p. 38). (d) Actual — 2022-23 Q2(b) (TF p. 48). (e) Actual — 2022-23 Q2(c) (TF p. 48). (f) Actual — 2023-24 Q3(a) (22 batch p. 2).

(a) A lot of 7 components (4 good, 3 defective) is sampled; 3 are taken. Find the expected number of good components in the sample. (15)

(b) X has density $f(x) = e^{-x}$, $x > 0$. Find the expected value of $g(X) = e^{2X/3}$. (18)

(c) Hospital stay $Y = X + 4$ days, where $f(x) = 32/(x+4)^3$, $x > 0$. Find the average stay. (10)

(d) Clearance time $Y = 3X - 2$ minutes, where $f(x) = (1/4)e^{-x/4}$, $x > 0$. Find the mean and variance of Y. (10)

(e) Compute $P(\mu - 2\sigma < X < \mu + 2\sigma)$ for $f(x) = 6x(1-x)$, $0 < x < 1$, and compare with Chebyshev's theorem. (10)

(f) State Chebyshev's theorem. 1000 applicants sit a test for 70 jobs; mean 60, SD 6, distribution symmetric about the mean. Can a person who scores 84 count on getting a job? (11)

✓ Solution

(a) X = number of good components; hypergeometric $f(x) = {}^4C_x \cdot {}^3C_{(3-x)} / {}^7C_3$, ${}^7C_3 = 35$.

```
x     :   0      1       2       3
f(x)  :  1/35  12/35   18/35   4/35
E(X) = 0(1/35) + 1(12/35) + 2(18/35) + 3(4/35) = (12 + 36 + 12)/35 = 60/35 = 12/7 ≈ 1.714
```

Check with the hypergeometric mean $nM/N = 3 \times 4/7 = 12/7$ ✓.

(b)

$$\mathbf{E(e^{2X/3}) = \int_0^\infty e^{2x/3}\, e^{-x}\, dx = \int_0^\infty e^{-x/3}\, dx = [-3e^{-x/3}]_0^\infty = 3}$$

(c) First check $\int_0^\infty 32(x+4)^{-3}\, dx = [-16(x+4)^{-2}]_0^\infty = 16/16 = 1$ ✓.

```
E(X) = ∫0^∞ x · 32/(x + 4)³ dx,  put u = x + 4:
     = 32 ∫4^∞ (u − 4)/u³ du = 32 [ −1/u + 2/u² ]4^∞ = 32 (1/4 − 2/16) = 32 × 1/8 = 4
E(Y) = E(X) + 4 = 8 days
```

(d) X is exponential with mean 4: $E(X) = \int x(1/4)e^{-x/4}dx = 4$, $E(X^2) = 2 \times 4^2 = 32$, $V(X) = 32 - 16 = 16$.

```
E(Y) = 3E(X) − 2 = 12 − 2 = 10 minutes
V(Y) = 3² V(X) = 9 × 16 = 144  (σ_Y = 12 minutes)
```

(e)

```
μ   = ∫0^1 x·6x(1 − x) dx = 6(1/3 − 1/4) = 1/2
E(X²) = ∫0^1 x²·6x(1 − x) dx = 6(1/4 − 1/5) = 3/10,   σ² = 3/10 − 1/4 = 1/20,  σ = 0.2236
μ ± 2σ = 0.5 ± 0.4472 → (0.0528, 0.9472)  (inside (0, 1))
P = ∫ 6x(1 − x) dx = [3x² − 2x³] over (0.0528, 0.9472) = 11√5/25 = 0.9839
Chebyshev (k = 2): P ≥ 1 − 1/4 = 0.75
```

The exact probability 0.984 is far above the Chebyshev lower bound 0.75 — the theorem is true but conservative because it uses no information about the shape of the distribution.

(f) Chebyshev: for any $k > 0$, $P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - 1/k^2$. Here $84 = 60 + 4 \times 6$, so $k = 4$.

```
P(36 < X < 84) ≥ 1 − 1/16 = 15/16  →  P(|X − 60| ≥ 24) ≤ 1/16
Symmetric about the mean → P(X ≥ 84) ≤ (1/2)(1/16) = 1/32
Number of applicants scoring 84 or more ≤ 1000/32 = 31.25, i.e. at most 31 people
```

Since at most about 31 of the 1000 applicants can score 84 or above and there are 70 jobs, a person scoring 84 is within the top 70 and can count on getting a job.

Quick Notes — Random Variables

Revision points for Lessons 7–9.

L7 · pmf

- Random variable = numerical outcome of chance; discrete (countable) or continuous (interval).
- pmf: $f(x) \geq 0$ and $\Sigma f(x) = 1$; solve $\Sigma = 1$ for k (k-table: $k = 1/10$; $\alpha(3/4)^x$: $\alpha = 1/4$).

L8 · pdf, CDF, Exponential

- pdf: $f \geq 0$, $\int f = 1$, $P(a < X < b) = \int_a^b f$; $P(X = a) = 0$.
- CDF $F(x) = \int_{-\infty}^{x} f$; $f = F'$; $P(a < X < b) = F(b) - F(a)$; carry earlier pieces' area.
- Exponential: $F = 1 - e^{-\lambda x}$, mean $1/\lambda$, memoryless; convert λ to the units of t.
- Fire station: 0.0186, 0.0186, $e^{-0.75} = 0.4724$, $e^{-3} = 0.0498$, x = 1.4 h.

L9 · Expectation, Chebyshev

- $E(X) = \Sigma xf$ or $\int xf$; $V = E(X^2) - \mu^2$; $E(aX + b) = aE + b$; $V(aX + b) = a^2V$.
- $E(e^{2X/3})$ for e^{-x}: 3. Hospital stay: $E(X) = 4$, $E(Y) = 8$. $Y = 3X - 2$: 10 and 144.
- Chebyshev: $P(|X - \mu| < k\sigma) \geq 1 - 1/k^2$; $6x(1 - x)$: exact 0.9839 vs 0.75.
- Jobs: $84 = \mu + 4\sigma \rightarrow$ at most 1/32 of 1000 ≈ 31 above → yes, gets a job.